VITREOUS HEMORRHAGE AFTER PLAQUE RADIOTHERAPY FOR UVEAL MELANOMA

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Purpose: The purpose of this article is to determine the incidence, etiology, management, and outcome of vitreous hemorrhage (VH) after plaque radiotherapy for uveal melanoma.

Methods: Retrospective review of medical records.

Results: Of 3,707 eyes treated with plaque radiotherapy for uveal melanoma, VH developed in 4.1% at 1 year, 15.1% at 5 years, and 18.6% at 10 years by Kaplan–Meier analysis. Presumed causes of VH included tumor necrosis (29%), proliferative radiation retinopathy (24%), posterior vitreous detachment (16%), vascular occlusion (5%), and unknown (19%). Tumor necrosis was the most common cause of VH early in the follow-up period (3% at 1 year), while proliferative radiation retinopathy was the most common source of VH later (6.2% at 15 years). The most common initial management was conservative observation for resolution in 48%, laser photocoagulation in 24%, and vitrectomy in 18%. After a mean follow-up period of 5 years, the VH was completely resolved in 41%, partially resolved in 19%, unresolved in 20%, worsened in 5%, and enucleation was necessary in 15%. By multivariable analysis, risk factors for development of VH were the presence of diabetic retinopathy at first visit (relative risk, 6.64), shorter tumor distance to the optic disc (relative risk, 1.07), greater initial tumor thickness (relative risk, 1.1), and break in the Bruch membrane (relative risk, 2.93). The rates of local tumor recurrence, extraocular extension, and distant metastasis in 74 patients who underwent vitrectomy for VH removal after tumor regression were similar to those in patients who did not have vitrectomy for VH.

Conclusion: Vitreous hemorrhage occurs after plaque radiotherapy for uveal melanoma in 15.1% of the patients by 5 years. The main factors predictive of VH included underlying diabetic retinopathy, closer tumor proximity to the disc, greater tumor thickness, and break in the Bruch membrane. After tumor regression, vitrectomy for blood removal appears to be safe.

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Vitreous hemorrhage (VH) is the extravasation of blood within the vitreous cavity and can occur spontaneously or after accidental or nonaccidental ocular trauma. In an analysis of 1,562 cases of spontaneous VH from 6 different studies, the most common causes were found to be proliferative diabetic retinopathy (32%), retinal tear (30%), retinal vein occlusion (11%), posterior vitreous detachment (PVD) without tear (8%), and proliferative sickle retinopathy, age-related macular degeneration, and macroaneurysm (2% each).[1–3]

Vitreous hemorrhage can occur after plaque radiotherapy for uveal melanoma, and the rates have been reported to range within 8% to 36% by 5 years of follow-up.[4–18] Previous reports of ocular complications after plaque radiotherapy have indicated VH as one of the less common radiation side effects, but a specific study on this entity is lacking in the ophthalmic literature. This finding can lead to temporary or permanent blindness, but more importantly, it can camouflage underlying retinal problems that might necessitate therapy and could

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interfere with adequate monitoring of the treated melanoma. We report herein the causes and management of VH after plaque radiotherapy for uveal melanoma in a large cohort of patients.

Patients and Methods

The medical records of all patients with uveal melanoma managed with plaque radiotherapy from June 1, 1978, to April 30, 2007, in the Ocular Oncology Service at the Wills Eye Institute, Philadelphia, PA, were reviewed. Patients who had developed vitreous and/or subhyaloid hemorrhage (VH) after plaque radiotherapy were included in the study. Those who did not have VH were included as control patients.

Baseline patient data included age, race, sex, history of diabetes mellitus, systemic hypertension, and hypercholesterolemia. Ocular findings recorded at the first visit included initial Snellen best-corrected visual acuity, intraocular pressure, and the presence of diabetic retinopathy.

Tumor data consisted of tumor tissue (choroid, ciliochoroid, ciliary body, iridociliary, iris, or iridociliochoroidal), distance of the posterior margin of the tumor to the optic disc and foveola, maximum tumor diameter (based on indirect ophthalmoscopy and B-scan ultrasonography), initial thickness (based on A- and B-scan ultrasonography for tumors involving the choroid and/or ciliary body and ultrasound biomicroscopy for tumors involving the iris), shape (dome, mushroom, or diffuse based on indirect ophthalmoscopy and B-scan ultrasonography), associated features (Bruch membrane rupture, retinal invasion, extraocular extension of the tumor, and orange pigment based on indirect ophthalmoscopy and B-scan ultrasonography), and the presence of subretinal fluid (based on indirect ophthalmoscopy and A- and B-scan ultrasonography).

Plaque radiotherapy data included radioactive isotope (iodine 125, ruthenium 106, cobalt 60, and iridium 192); plaque shape (round or notched); plaque size; hours of radiation; radiation doses to the apex, base, optic nerve, fovea, and lens; and radiation rates to the apex, base, optic nerve, fovea, and lens. Information on transpupillary thermotherapy, if performed, included number of sessions.

The follow-up examinations were generally made at 4-month to 6-month intervals up to 5 years and at 6-month to 12-month intervals thereafter. The development of VH was determined using indirect ophthalmoscopy, fundus photography, and B-scan ultrasonography. In each case, an attempt was made to determine the etiology of VH (tumor necrosis, PVD, retinal invasion, vascular occlusion, proliferative radiation retinopathy [PRR], proliferative diabetic retinopathy, tumor recurrence, Valsalva maneuver, or the previous use of Coumadin or diagnostic fine needle aspiration biopsy). The association with diabetic retinopathy and retinal vascular occlusion was also recorded. Diagnosis of PRR was based on the presence of retinal (NVE) or optic disc neovascularization (NVD). Tumor necrosis was defined as acute ischemic shrinkage of the melanoma with associated pigment dispersion and rupture of blood vessels within the tumor that cause the release of blood into the vitreous.

The different treatment modalities for management of VH were recorded (observation, panretinal photocoagulation, vitrectomy, enucleation, cryotherapy, and intravitreal triamcinolone), as were the number of episodes of VH. Final outcomes for the VH were graded as no change (stable), partially resolved, completely resolved, or worsened. Other radiation side effects recorded included PRR, non-PRR, tumor exudation, radiation maculopathy, radiation papillopathy, radiation-related scleral necrosis, glaucoma, and cataract.

At the last follow-up visit, the following information was recorded: best-corrected visual acuity, patient status (alive and well, alive with metastasis, dead of metastasis, dead of other cause, and lost to follow-up), and globe status (enucleation). The visual status was assessed by Snellen visual acuity and considered stable when the change in vision was <3 lines difference, moderate loss when loss of ≥ 3 lines of visual acuity was recorded, and severe loss in visual acuity when vision was worse than 5/200.[5] Improvement in visual acuity was defined as change in vision to at least one half of the initial visual angle.

Statistical Analysis

Comparison of patient demographic features, tumor features, plaque radiotherapy parameters and associated treatments and procedures before VH, and radiation-associated side effects among those who developed VH (cases) after plaque and those did not develop VH (control patients) was performed using independent samples *t*-test/Wilcoxon rank-sum test, where appropriate, for continuous variables and Fisher exact test/chi-square test for categorical variables.

Kaplan–Meier survival analyses were performed to estimate the overall probability of development of VH, specific relationship of VH secondary to tumor necrosis, VH secondary to PVD, and VH secondary to PRR at different time intervals. Univariable and multivariable Cox regression analyses were performed for the factors predictive of VH. The risk factor analyses were also performed for a subgroup of those with interval between plaque and VH of ≤ 1 year and separately

Table 1. Vitreous Hemorrhage After Plaque Radiotherapy for Uveal Melanoma: Patient Demographics

Features	VH, n = 403	Control, n = 3,304	*P**
Age, years			
Median (mean, range)	60 (59, 12–88)	59 (59, 15–98)	0.863†
Race, n (%)			0.292‡
White	391 (97)	3,240 (98)	
Black	3 (<1)	20 (<1)	
Hispanic	8 (2)	32 (<1)	
Asian	1 (<1)	12 (<1)	
Sex, n (%)			0.082
Male	221 (55)	1,656 (50)	
Female	182 (45)	1,648 (50)	
Medical history§, n (%)			
Diabetes	62 (15)	287 (9)	<0.001
Hypertension	142 (35)	751 (23)	<0.001
Hypercholesterolemia	23 (6)	127 (4)	0.081

*Fisher exact test.
†Independent samples *t*-test.
‡Chi-square test.
§Some patients had >1 condition.

for a subgroup of those with interval >1 year. The hazard ratios for all variables were calculated as discrete variables except for patient age at presentation, distance to the optic disc and foveola, largest base, thickness, plaque size, and radiation-associated parameters. These were evaluated instead as continuous variables. Multivariate Cox regression models included variables significant on a univariable Cox regression model at level

Table 2. Vitreous Hemorrhage After Plaque Radiotherapy for Uveal Melanoma: Tumor Features

Features	VH, n = 403	Control, n = 3,304	*P**
Tumor tissue, n (%)			0.097†
Choroid	351 (87)	2,811 (85)	0.298
Ciliochoroid	46 (11)	408 (12)	0.630
Ciliary body	1 (<1)	47 (1)	—
Iridociliary	1 (<1)	24 (<1)	—
Iris	0 (0)	0 (0)	—
Iridociliochoroidal	4 (<1)	14 (<1)	—
Distance to optic nerve, mm			
Median (mean, range)	3 (3, 0–15)	4 (5, 0–20)	<0.001‡
Distance to foveola, mm			
Median (mean, range)	3 (3, 0–17)	3.5 (4.8, 0–22)	<0.001‡
Largest base, mm			
Median (mean, range)	11 (11, 4–19)	11 (10.7, 2–22)	0.294§
Thickness, mm			
Median (mean, range)	5 (6, 1–14)	4 (4.8, 0.25–15.5)	<0.001§
Shape, n (%)			<0.001†
Dome	219 (54)	2,662 (81)	
Mushroom	160 (40)	410 (12)	
Diffuse	24 (6)	232 (7)	
Associated features, n (%)			
Bruch membrane rupture	194 (48)	410 (12)	<0.001
Retinal invasion	74 (18)	220 (7)	<0.001
Extraocular extension of the tumor	5 (1)	50 (2)	0.829
Orange pigment	131 (32)	—	—
Presence of subretinal fluid, n (%)			<0.001
No	65 (16)	827 (25)	
Yes	338 (84)	2,477 (75)	

*Fisher exact test.
†Chi-square test.
‡Wilcoxon rank-sum test.
§Independent samples *t*-test.

Table 3. Vitreous Hemorrhage After Plaque Radiotherapy for Uveal Melanoma: Plaque Radiotherapy Parameters

Features	VH, n = 403	Control, n = 3,304	*P**
Isotope, n (%)			<0.001†
Iodine 125	387 (96)	2,786 (84)	
Ruthenium 106	1 (<1)	64 (2)	
Cobalt 60	13 (3)	346 (10)	
Iridium 192	2 (<1)	108 (3)	
Plaque shape, n (%)			<0.001‡
Round	256 (64)	2,610 (79)	
Notched	147 (36)	694 (21)	
Plaque size, mm			
Median (mean, range)	15 (16, 10–22)	15 (15.5, 8–25)	<0.001§
Hours of radiation			
Median (mean, range)	97 (104, 22–275)	97 (115, 26–511)	0.019
Radiation dose to, Gy			
Median (mean, range)			
Apex	82 (88, 45–156)	83 (88, 12–300)	0.536
Base	313 (319, 108–815)	300 (289, 26–815)	<0.001
Disc	55 (75, 4–441)	33 (46, <1–878)	<0.001
Foveola	58 (91, 3–656)	38 (64, <1–23)	<0.001
Lens	19 (29, <1–211)	17 (27, <1–488)	0.022
Radiation rate to, cGy/hour			
Median (mean, range)			
Apex	86 (89, 29–441)	85.3 (85.1, 6.4–1,088.8)	0.003
Base	316 (315, 34–852)	258.6 (268, 14–1,475)	<0.001
Disc	57 (77, 6–413)	34 (52.5, 0.48–5,503)	<0.001
Foveola	63 (95, 0–463)	40 (64.2, 0.96–528)	<0.001
Lens	18 (27, 3–212)	15.9 (30, 0.2–3,044)	0.002

*Wilcoxon rank-sum test.
†Chi-square test.
‡Fisher exact test.
§Independent samples *t*-test.

($P < 0.05$) to identify the combination of factors best related to the outcome of VH.

Results

Between June 1978 and April 2007, 3,707 patients (3,707 eyes) with uveal melanoma received treatment with plaque radiotherapy, of which 403 patients (403 eyes) (10.6%) developed VH during subsequent follow-up. Patients who did not develop VH were included as control patients. Table 1 shows the comparative demographic features of the VH group and control group. There was no difference in age, race, and sex. However, the VH group was more likely to have underlying diabetes ($P < 0.001$) and hypertension ($P < 0.001$). The visual acuity at presentation was between 20/20 and 20/40 in 260 patients (64%), between 20/50 and 20/100 in 87 (22%), and 20/200 and worse in 56 (14%). Median intraocular pressure was 15 mmHg (mean, 15 mmHg; range, 5–28 mmHg). Non–proliferative diabetic retinopathy was present in 11 patients (3%) at the first visit, while proliferative diabetic retinopathy was found in 1 patient (<1%).

A comparative analysis of tumor features (Table 2) in the 2 groups showed no difference in tumor location, basal diameter, or extraocular extension. Nevertheless, the VH group was more likely ($P < 0.0001$) to have tumors located closer to the optic disc and fovea, with greater thickness, higher rate of break in the Bruch membrane, and higher rate of retinal invasion. A comparative analysis of radiation parameters (Table 3) in the 2 groups showed that the VH group was more likely ($P < 0.001$) to have received a notched plaque and a higher radiation dose and radiation rate to the tumor base, the foveola, and the optic disc. After plaque radiotherapy, transpupillary thermotherapy was delivered in 121 patients (30%) (mean, 1 session per patient).

The main cause for VH included tumor necrosis in 117 patients (29%), PRR in 98 (24%), PVD in 64 (16%), and retinal vascular occlusion in 21 (5%) (Figure 1). Other less common causes included retinal invasion in 12 patients (3%), tumor recurrence in 9 (2%), the use of Coumadin in 2 (<1%), Valsalva maneuver in 2 (<1%), proliferative diabetic retinopathy in 1 (<1%), and tumor fine needle aspiration biopsy in

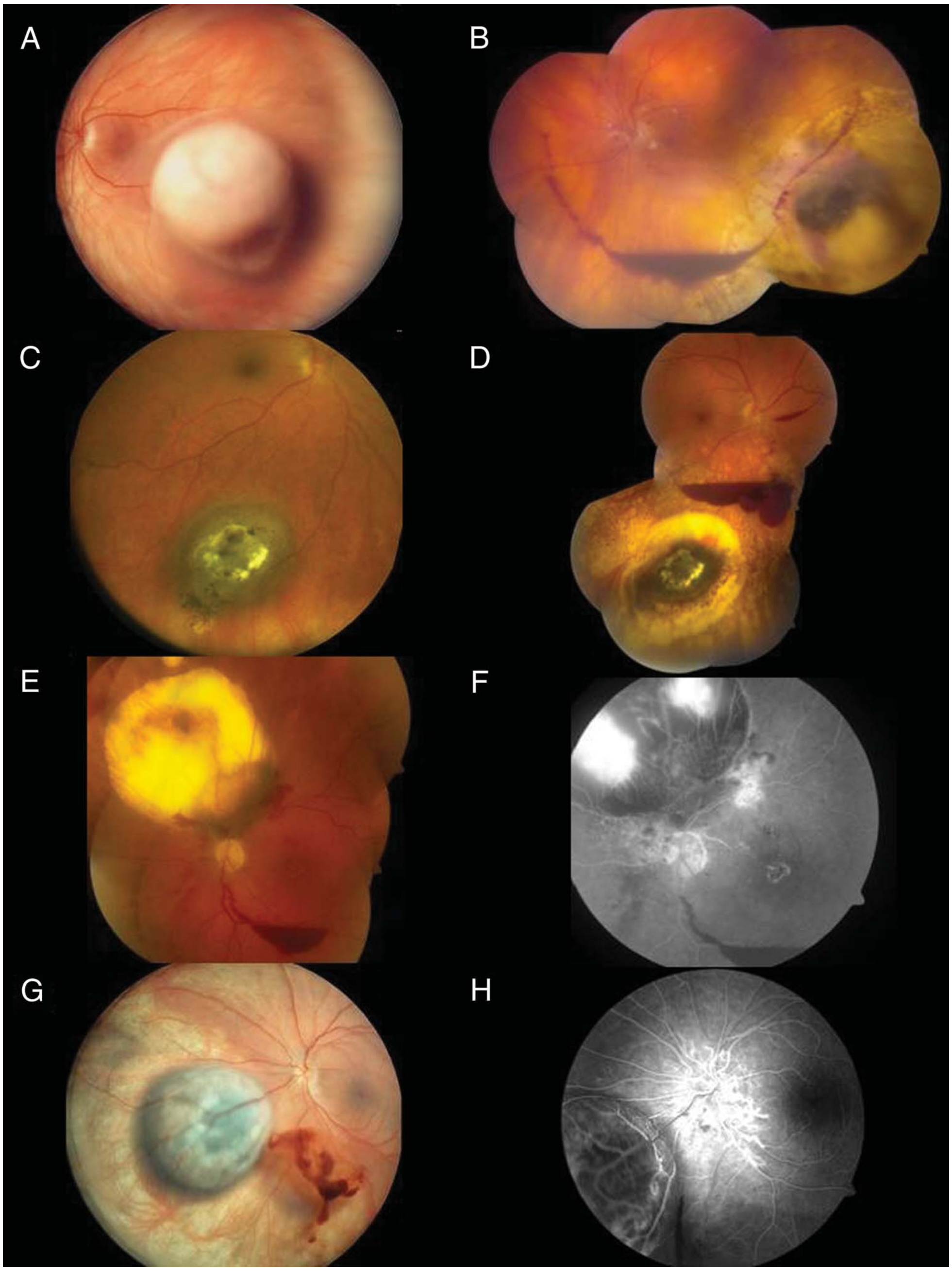

Fig. 1. An 80-year-old woman presented with an amelanotic choroidal melanoma (**A**) that was treated with plaque radiotherapy and developed subhyaloid hemorrhage (**B**) 3 years later from PVD. A 46-year-old man with a choroidal melanoma (**C**), treated with plaque radiotherapy, showed subhyaloid hemorrhage from retinal neovascularization 28 months later. A 56-year-old woman with an irradiated regressed choroidal melanoma. (**D**) 28 months later. (**E**) developed preretinal hemorrhage inferiorly at 26-month follow-up from retinal neovascularization (**F**). A 55-year-old man with an irradiated regressed choroidal melanoma (**G**) developed VH from disc neovascularization (**H**) 2 years after treatment.

1 (<1%). In 76 patients (19%), no definite etiologies were identified because of a poor view of the fundus from VH in 58 patients (14%) or from cataract in 18 patients (4%). At 5 years, VH developed in 15.1% of patients according to Kaplan–Meier analysis (Table 4). Table 4 shows the incidence of VH for patients with tumor necrosis, PVD, and PRR as well (Figures 2–5).

Proliferative radiation retinopathy was diagnosed in the form of NVE in 56 patients (14%), NVD in 36 (9%), and a combination of both in 19 (5%). Change in best-corrected visual acuity after VH was measured as stable in 126 patients (31%), moderate loss in 62 (15%), and severe loss in 215 (53%). The number of episodes of VH registered was 1 in 358 patients (89%), 2 in 41 (10%), and 3 in 4 (1%). Management of VH consisted most frequently of observation in 195 patients (48%), panretinal photocoagulation in 97 (24%), vitrectomy in 74 (18%), and enucleation in 62 (15%). Other less commonly used treatments included cryotherapy and intravitreal triamcinolone in 1 patient each (<1%). The treatment outcomes at the last visit were unresolved VH in

Table 4. Vitreous Hemorrhage After Plaque Radiotherapy for Uveal Melanoma: Time to Onset of VH by Kaplan–Meier Analysis

Outcomes	1 Year (%)	2 Years (%)	3 Years (%)	5 Years (%)	10 Years (%)	15 Years (%)
VH	4.1	7.8	11.5	15.1	18.6	20.3
VH secondary to						
Tumor necrosis	3.0	3.7	3.9	4.1	4.2	4.2
PVD	0.45	1.2	2.0	2.9	3.8	3.8
PRR	0.28	1.5	2.9	4.8	5.5	6.2

80 (20%), partially resolved VH in 75 (19%), completely resolved VH in 166 (41%), and worsened VH in 20 (5%). Vitreous hemorrhage appeared in association with other radiation-induced side effects, such as non-PRR in 261 patients (65%), exudative retinopathy in 77 (19%), radiation maculopathy in 198 (49%), radiation papillopathy in 138 (34%), radiation cataract in 245 (61%), neovascular glaucoma in 107 (27%), and radiation scleral necrosis in 5 (1%).

The mean follow-up interval between the plaque brachytherapy application and the last recorded visit was 62 months (range, 2–301 months). Findings at the last visit included a visual acuity of 20/40 or better in 29 patients (7%), 20/100 to 20/50 in 36 (9%), and 20/200 or worse in 338 (84%). Systemic status of the patients at the last visit showed that 365 patients (91%) were alive and well and 31 (8%) were alive with metastasis. Five patients (1%) had died of metastasis, and 2 (<1%) were lost to follow-up. Enucleation was necessary in 62 patients (15%). Tumor recurrence was found in 27 patients (7%) at the last visit. For the specific group of patients who underwent vitrectomy (n = 74), the outcome after a mean follow-up of 73 months (median, 59 months; range, 6–195 months) showed that the rate of enucleation, distant metastasis, and tumor recurrence was 9% (7 patients), 5% (4 patients), and 3% (2 patients), respectively. The final outcome after treatment showed that 72% of patients having undergone vitrectomy showed complete resolution of the VH compared with 55% of those receiving panretinal photocoagulation and 35% of those managed with observation.

On multivariate analysis, significant risk factors for development of VH were the presence of diabetic retinopathy at first visit (relative risk, 6.64; 95% confidence interval, 3.62–12.18), shorter tumor distance to the optic disc (relative risk, 1.07; 95% confidence interval, 1.04–1.11), greater initial tumor thickness (relative risk, 1.1; 95% confidence interval, 1.04–1.16), and break in the Bruch membrane (relative risk, 2.93; 95% confidence interval, 2.24–3.82), as shown in Table 5.

Discussion

Vitreous hemorrhage in the general population can be caused by several pathologic mechanisms.[2,3] The

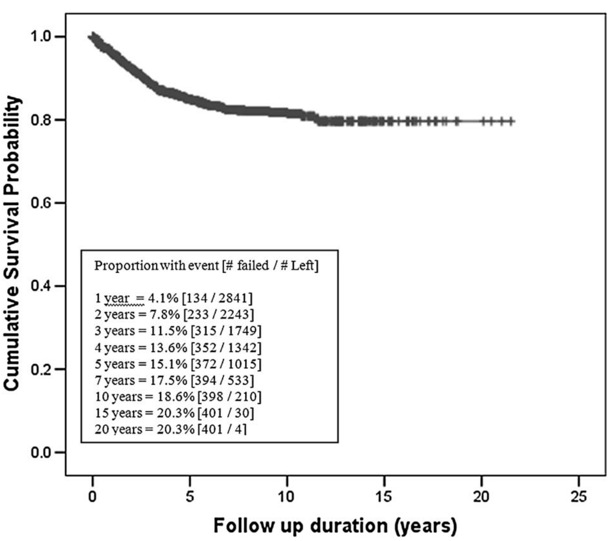

Fig. 2. Time to onset of VH according to Kaplan–Meier analysis.

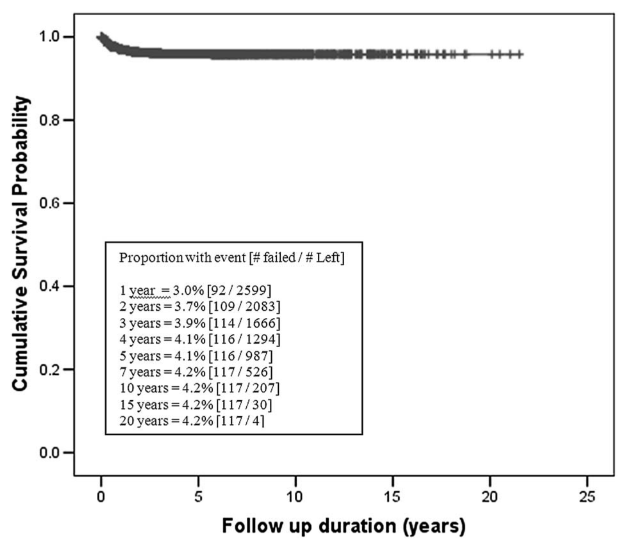

Fig. 3. Time to onset of VH secondary to tumor necrosis according to Kaplan–Meier analysis.

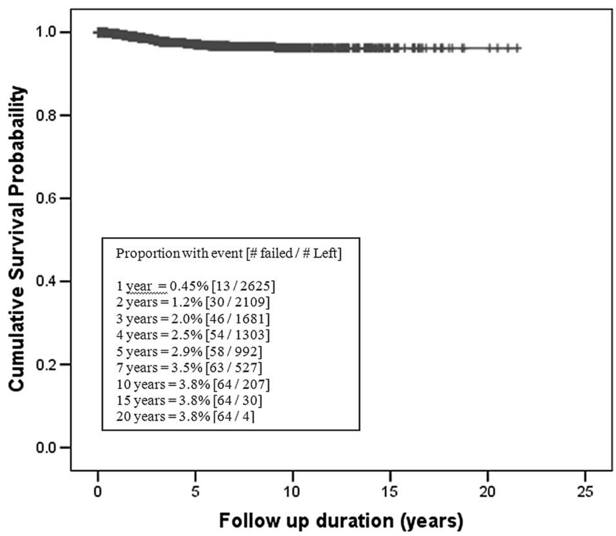

Fig. 4. Time to onset of VH secondary to PVD according to Kaplan–Meier analysis.

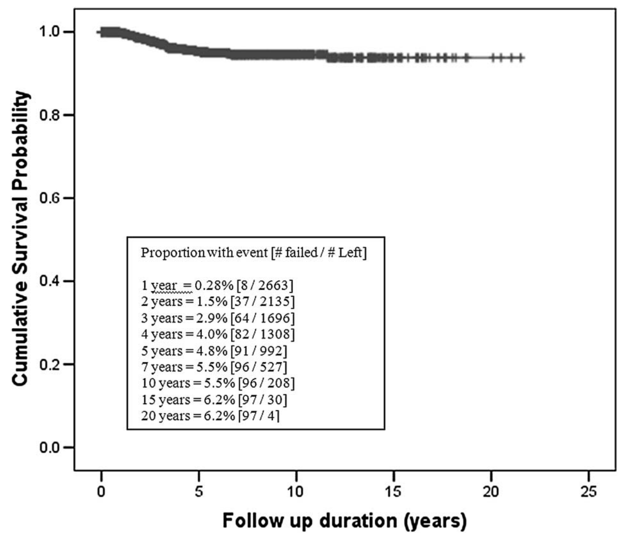

Fig. 5. Time to onset of VH secondary to PRR according to Kaplan–Meier analysis.

most common is retinal ischemia with subsequent release of proangiogenic factors that results in the formation of new fragile retinal vessels, termed "neovascularization," as seen in proliferative diabetic retinopathy, sickle retinopathy, retinal vein occlusion, or retinopathy of prematurity.[2,3] A second process that results in VH is the rupture of a normal retinal vessel from a retinal tear caused either by a retinal break or by PVD.[2,3] Less frequently, VH can be caused by subretinal bleeding with secondary extravasation into the vitreous cavity, such as in age-related macular degeneration, choroidal melanoma, or Terson syndrome.[2,3] In eyes treated with radiotherapy for melanoma, the source of VH can range from ischemic processes to PVD or to treatment-induced tumor necrosis. The most common presumed etiologies identified for VH in the present study were tumor necrosis in 117 cases (29%), PRR in 98 (24%), and PVD in 64 (16%), in concordance with previous reports on nonmelanoma patients.[2,3] In the early postoperative period (within 12 months of treatment), tumor necrosis was the most common causative factor for VH (3% at 1 year vs. <1% at 1 year for patients with PVD and PRR, respectively). In the late postoperative period (from 10 to 15 years), the most common cause for VH was PRR (6.2% at 15 years vs. 4.2 and 3.8% at 15 years for tumor necrosis and PVD, respectively) (Table 4).

In the current study, Kaplan–Meier analysis showed that the incidence of VH after plaque radiotherapy for uveal melanoma was 15.1% at 5 years and 18.6% at 10 years. Gunduz et al[13] described an incidence of VH of 7.7% at 5 years of follow-up in 1,300 patients treated with the same modality. Shields et al[14] found that the incidence of VH after plaque radiotherapy in 354 patients with large melanomas (≥8 mm in thickness) was 23% at 5-year follow-up. Similarly, Puusaari et al[15] found that the rate of VH in 96 patients with large melanoma treated with plaque radiotherapy was 36% at 5 years.

In the present study, diabetes, diabetic retinopathy, hypertension, and hypercholesterolemia were found to be significant risk factors for the development of VH in the univariable analysis. Diabetic retinopathy was found to be a risk factor in the multivariable analysis as well. Similarly, previous studies have proved diabetes to be a risk factor for the development of PRR.[19]

The most important tumor features related to the development of VH in the multivariable analysis were the presence of break in the Bruch membrane, greater tumor thickness, and shorter tumor distance to the optic disc (Table 5). In a previous report on plaque radiotherapy for large melanoma, the Bruch membrane rupture was found in 32%, 23% of which developed VH at 5 years after radiotherapy.[14] Similarly, a previous report on PRR after plaque radiotherapy for uveal melanoma also revealed shorter tumor distance to the optic disc as a significant risk factor for the development of VH.[19]

In terms of management, vitrectomy had the highest therapeutic efficacy, with complete resolution of the VH in 53 cases (72%). The vitrectomy was performed after a period of observation in 47 cases at an interval of 2 months to 53 months (median, 8 months) after the VH was discovered. This observation period was to allow potential spontaneous resolution of the blood. More than any other therapy, vitrectomy leads to the highest rate of VH resolution in 72%. An important additional finding in this study was the fact

Table 5. Vitreous Hemorrhage After Plaque Radiotherapy for Uveal Melanoma: Factors Predictive of VH

Variables	Control, n = 3,304	VH, n = 403	*P*	Relative Risk	95% Confidence Interval
Univariate analysis					
Race, n (%)					
Hispanic vs. white*	32 (1.0)	8 (2.0)	0.002	2.96	1.47–5.96
Sex, n (%)					
Male vs. female*	1,656 (50.1)	221 (54.8)	0.015	1.28	1.05–1.55
Patient history, n (%)					
Diabetes vs. no diabetes*	287 (8.7)	62 (15.4)	<0.001	1.98	1.51–2.60
Hypertension vs. no hypertension*	751 (22.7)	142 (35.2)	<0.001	2.17	1.77–2.66
Hypercholesterolemia vs. no hypercholesterolemia*	127 (3.8)	23 (5.7)	0.001	2.01	1.32–3.06
Visual acuity, n (%)					
20/50 to 20/200 vs. 20/20 to 20/40*	782 (23.7)	111 (27.5)	0.011	1.33	1.07–1.67
Diabetic retinopathy, n (%)					
Yes vs. no*	14 (0.4)	12 (3.0)	<0.001	10.39	5.83–18.54
Distance to optic nerve (median)	4.0	2.5	<0.001	1.16a	1.12–1.20
Distance to foveola (median)	3.5	3.0	<0.001	1.09a	1.06–1.13
Largest base (mean)	10.7	10.9	0.017	1.04c	1.01–1.07
Thickness (mean)	4.8	5.8	<0.001	1.18c	1.14–1.22
Shape, n (%)					
Mushroom vs. dome*	410 (12.4)	160 (39.7)	<0.001	4.68	3.81–5.75
Bruch membrane rupture, n (%)					
Yes vs. no*	410 (12.4)	194 (48.1)	<0.001	5.94	4.88–7.22
Retinal invasion, n (%)					
Yes vs. no*	220 (6.7)	74 (18.4)	<0.001	3.33	2.59–4.29
Subretinal fluid, n (%)					
Yes vs. no*	2,477 (75.0)	338 (83.9)	<0.001	1.83	1.40–2.39
Plaque shape, n (%)					
Notched vs. round*	694 (21.1)	147 (36.5)	<0.001	2.10	1.71–2.57
Plaque size (mean)	15.5	16.1	<0.001	1.14c	1.09–1.18
Radiation rate apex (mean)	85.1	88.6	0.025	1.02f	1.002–1.03
Radiation base (mean)	28,866.8	31,680.3	<0.001	1.02g	1.01–1.03
Radiation rate base (mean)	268.1	315.1	<0.001	1.29h	1.21–1.37
Radiation disk (median)	3,271.7	5,496.0	<0.001	1.05g	1.04–1.06
Radiation foveola (median)	3,767.0	5,791.3	<0.001	1.11i	1.05–1.16
Radiation rate foveola (median)	40.0	63.3	<0.001	1.55h	1.38–1.74
Multivariate analysis†					
Diabetic retinopathy					
Yes vs. no*	—	—	<0.001	6.64	3.62–12.18
Distance to optic nerve	—	—	<0.001	1.07a	1.04–1.11
Thickness	—	—	0.001	1.10c	1.04–1.16
Bruch membrane rupture					
Yes vs. no*	—	—	<0.001	2.93	2.24–3.82

The online alphabets refer to the following: a, per 1-mm decrease; b, per 10% increase; c, per 1-mm increase; d, per 10-hour decrease; e, per 1,000-cGy decrease; f, per 10-cGy/hour increase; g, per 1,000-cGy increase; h, per 100-cGy/hour increase; k, per 10,000-cGy increase; j, per 1-session increase; and l, per 1-year decrease.

*Reference variable.

†Considering all significant variables by univariate analysis into the model.

that vitrectomy did not increase the rate of tumor recurrence (3%) or distant metastasis (5%) at 5-year follow-up compared with patients who did not undergo vitrectomy (7 and 9% at 5 years, respectively). In one small case series of nine patients with irradiated uveal melanoma who underwent vitrectomy for VH removal, Foster et al[20] described one patient who had subsequent intraocular dissemination of melanoma cells.

In conclusion, VH develops after plaque radiotherapy for uveal melanoma in 15.1% of patients at 5 years. Risk factors for VH include diabetic retinopathy, shorter tumor distance to the optic disc, greater tumor thickness, and the presence of break in the Bruch membrane. The most common presumed etiologies included tumor necrosis (29%), PRR (24%), and PVD (16%). Pars plana vitrectomy was the most effective treatment option for visual rehabilitation,

and it provided complete stable resolution of hemorrhage in 72% of cases, with no increased risk for tumor recurrence or metastasis.

Key words: eye, tumor, choroid, melanoma, plaque radiotherapy, complication, vitreous hemorrhage.

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